Capstone General Instructions Guide

Everything you need to know before you begin

30-Day Duration	100 Marks Total	Expert Evaluation
From enrollment date	Min 75 to pass	Live viva session

Introduction to the Capstone

The Capstone is the final milestone of your program - your chance to apply everything you've learned to a real-world problem, end to end. Unlike module exercises, it is open-ended. You are expected to make deliberate choices, experiment, and justify your decisions. The quality of your thinking matters as much as the final output.

What the Capstone Consists Of:

1. Problem Statement -The business context and the exact problem you need to solve. Read this before writing a single line of code.

2. Dataset - A labelled training set, an unlabelled test set, and a sample submission file. Build and validate your pipeline on the training data, then generate final predictions for the test set.

3. Suggested Approach & Technical Steps - A recommended end-to-end pipeline covering data loading, preprocessing, feature engineering, model building, and evaluation. These are guidelines, not rules — cover all core steps but feel free to go beyond them.

5. Deliverables - You must submit a Code File (notebook or script, runs end-to-end without errors), a Presentation (10–15 slides), and a Predictions File (CSV matching the sample submission format exactly).

6. Evaluation Rubrics - Your work is scored out of 100: 60 marks for the Project Component (data prep, modeling, code quality) and 40 marks for Technical Understanding, assessed live during your viva.

7. Tips & Common Mistakes - A curated list of best practices and frequent errors. Worth reading before you start and again before you submit.

8. Submission Checklist - A final check before uploading. Go through it line by line — it catches the most common gaps.

How You Will Be Evaluated

After uploading your submission, you book a **mandatory live evaluation session** with an Expert. The session has two parts:

- **Solution Review** — The Expert reviews your notebook, slides, and predictions file, and asks you to walk through and explain the choices you made.
- **Viva** — You are asked conceptual questions from the broader program curriculum to assess your understanding of the techniques you used.

You need a combined score of **at least 75 out of 100** to pass. Results are communicated by email and on your Evaluation Dashboard.

1. Capstone Duration & Timeline

Your capstone window begins the moment you enroll. Plan your time carefully — late submissions are not accepted.

Detail	Information
Duration	30 days from the date of capstone enrollment
Submission Deadline	Must be submitted before the 30-day window closes
Capstone Evaluation Session	Book it immediately after submission of Capstone to avoid delays.
Final Walkthrough	Make sure to submit a Zip file – consisting of all the files in the format mentioned in each capstone. Also, do not forget to submit your PPT

⚠ **Important:** Do not wait until the last day. Build, test, and validate your solution early, so you have time to iterate before submission.

2. Final Submission & Evaluation Process

What to Submit

Deliverable	Details
Code File	Jupyter Notebook (.ipynb) or Python script — must run end-to-end without errors
Presentation	10–15 slide deck (PowerPoint or PDF) covering approach, experiments, results
Predictions File	Predictions.csv matching the sample_submission.csv format (where applicable)

Evaluation Steps

- Upload your completed solution to the platform.
- Book a mandatory evaluation session with an Expert.
- The Expert will review your solution in full.
- You will be asked questions based on your project and overall program learning.
- Your performance will be evaluated across the defined rubric metrics mentioned above

📌 **Note:** The evaluation is a live session. Be ready to explain every decision in your notebook — from preprocessing to final model selection.

3. Scoring & Certification

Item	Details
Total Marks	100
Passing Threshold	Minimum 75 marks required to clear the evaluation
Evaluation Basis	Project component (60 marks) + Technical understanding / Viva (40 marks)

Score Breakdown

- **Project Component (60 marks):** Problem understanding, data preparation, model building, code quality, and deliverable completeness.
- **Technical Understanding (40 marks):** Assessed during the live viva — Python proficiency, EDA, feature engineering, ML/DL/NLP concepts, and your ability to explain your choices.

4. Result

Your result will be communicated through email and will also be available on your Evaluation Dashboard.

✅ If You Pass	❌ If You Do Not Pass
• You will receive your final program certificate • You will also get an opportunity to appear for the WSU (Western State University) certification	• You will not be eligible for the final program certificate • You can reattempt the capstone by paying a fee of INR 3,000

5. Important Guidelines

This capstone is a real-world assessment of the skills you have built during the program. The following guidelines apply to all capstone projects.

Academic Integrity

- Work independently. Collaboration and plagiarism of any kind are strictly prohibited.
- Do not copy code, analysis, or writeups from other participants or online sources.

- Your notebook, slides, and predictions must entirely reflect your own work.

Approach & Mindset

- Approach the problem sincerely and thoughtfully — treat it like a real business assignment.
- Start simple. Establish a baseline before building complex models.
- Document every decision. Use markdown cells to explain your reasoning at each step.
- Revalidate your solution before final submission — ensure the notebook runs end-to-end without errors.

Presentation & Communication

- Clearly articulate your approach, experiments, and conclusions in both your notebook and slides.
- The slide structure provided in each capstone brief is a recommended format — adjust as needed, but cover all required sections.
- Be prepared to explain any part of your solution during the live evaluation session.

6. Technical Standards

Data Handling & Leakage Prevention

- **Don't use any information from the test set when preparing your data. All preprocessing steps should be done using only the training data.**
- During validation, these steps should be done only on the training data first, and then the same setup should be applied to the validation data.
- After final model selection, fit the full pipeline on the entire training set, then apply it to the test set.
- Keep metadata (source file, page number, unique IDs) intact throughout — it is essential for citations and submission formatting.

Model Development

- Always establish a simple baseline model before building complex ones.
- **Minimum scope:** train and compare at least 3 models; include a clear justification for the final selected model.
- Use stratified cross-validation or stratified splits for reliable evaluation.
- Report the main evaluation metric along with class-wise metrics like precision, recall, and F1 for each class. A high overall score can sometimes hide poor performance on smaller or less frequent classes.
- Version your experiments — maintain a table of model name, hyperparameters, and evaluation scores.

EDA & Visualisation

- **Minimum scope:** at least 5 meaningful EDA visualisations.

- Explore class distribution, feature patterns, text characteristics, and any dataset-specific nuances (e.g. drug distribution, paper coverage).
- Use your EDA findings to justify preprocessing and modeling choices downstream.

Code Quality

- Structure your notebook into clearly labelled sections that match the approach steps.
- Write readable, modular, and well-commented code — avoid redundant or dead code.
- Save trained models using joblib or pickle so you do not need to retrain from scratch.
- Validate that your predictions file has the correct number of rows, column names, and value types before submitting.

7. Universal Submission Checklist

Use this checklist before every capstone submission. Requirements specific to individual capstones take precedence — refer to your capstone brief for project-specific details.

Code

- Notebook or script runs end-to-end without errors (restart & run all)
- All sections clearly labelled
- At least 5 meaningful EDA visualisations included
- At least 3 models trained, compared, and evaluated
- Final model selected with clear written justification
- No data leakage — all preprocessing fit only on training data
- Reproducible results — random seeds set where applicable
- All experiments (models, hyperparameters, metrics) documented inline

Presentation

- 10–15 slides covering all required sections from the capstone brief
- Architecture diagram or pipeline overview included
- Embedding / feature comparison table documented
- Sample outputs with citations or predictions shown
- Challenges, learnings, and conclusion covered

Submission File (where required)

- File has exactly the required columns with correct names
- Row count matches the test set exactly
- Prediction values are valid (e.g. 0, 1, 2 — no floats or NaN)
- IDs match the test set exactly and are in the correct order

8. Frequently Asked Questions

Q Can I use any model or library?

A Yes. The capstone brief suggests models and libraries, but these are recommendations. You are encouraged to experiment. Justify every choice you make.

Q What if I miss the 30-day deadline?

A Late submissions cannot be accepted. Ensure you plan your timeline from day 1.

Q Can I attempt the reattempt immediately after failing?

A You may reattempt after paying the INR 3,000 fee. Contact your college mentor for the reattempt process.

Q What should I do if I am unclear about the problem statement?

A Re-read the capstone brief carefully, especially the Key Challenge and Data Description sections.

Good Luck!

This capstone is designed to reflect the quality and rigour of real-world Data Science and AI work.

Focus on understanding deeply, experimenting rigorously, and building something you are proud to present.